



Teyzix Core Internship (June Batch)

Task Title: Customer Behavior Analytics & Churn Prediction Dashboard

Task ID: DA-INT-1

Domain: Data Analytics

Difficulty Level: Intermediate

Assigned Date: 11 June 2026

Submission Deadline: 19 June 2026

Objective

Build an end-to-end system that analyzes customer behavior data, generates business insights, and predicts churn probability using machine learning models.

Business Problem Statement

Companies struggle to understand customer behavior patterns and identify early signs of churn. TEYZIX CORE requires a combined analytics and ML system that transforms raw customer data into actionable insights and predictions.

Task Requirements

1. Data Analysis Module

- Load dataset (CSV format)
- Perform exploratory data analysis
- Handle missing values and outliers
- Generate summary statistics

2. Feature Engineering

- Create behavioral features such as
 - Usage frequency
 - Tenure
 - Payment history patterns
 - Support interaction count
- Trend-based features

3. **Data Visualization**

- Revenue trends
- Customer segmentation charts
- Churn distribution
- Correlation heatmap
- Category-wise comparisons

4. **Machine Learning Model**

- Train at least 2 models
- Logistic Regression
- Random Forest / XGBoost

Evaluate using

- Accuracy
- Precision
- Recall
- F1 Score
- ROC-AUC

5. **Customer Segmentation**

- Segment customers into
- High value
- Medium value
- Low value

Based on behavior + spending patterns

6. **Basic Prediction System**

- Predict churn probability
- Assign risk category
- Low / Medium / High risk

7. **Business Insights Report**

- Top churn reasons
- High-risk customer traits
- Revenue impact estimation

Technical Requirements

- Python required
- Use pandas, numpy, matplotlib/seaborn
- Use scikit-learn for ML models
- Modular notebook or scripts required

Expected Deliverables

- Python notebook (.ipynb)
- Dataset used
- Visualizations

Model results report
Summary insights document

Evaluation Criteria

Data analysis 25%
Feature engineering 20%
Model performance 20%
Visualization quality 15%
Insights clarity 10%
Code structure 10%

Bonus Challenge

Streamlit dashboard
SHAP explainability
Automated weekly prediction pipeline
Email report generation